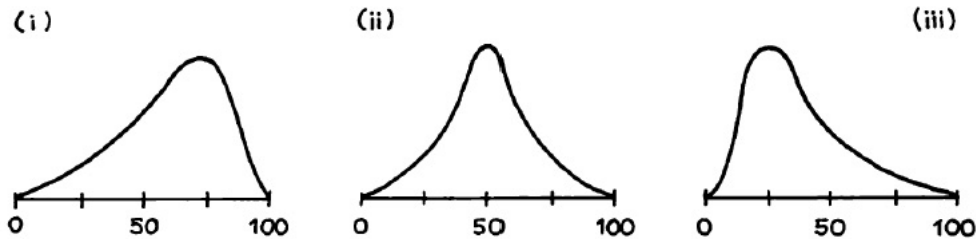


Quiz #1

1. Below are sketches of histograms for three variables.



- (a) In scrambled order, the means are 40, 50, 60. Match the histograms with the averages.

- (i) Average is 60
- (ii) Average is 50
- (iii) Average is 40

- (b) Match the histogram with the description:
- the median is smaller than the mean
 - the median is about equal to the mean
 - the median is larger than the mean

- (i) the median is larger than the mean
- (ii) the median is about equal to the mean
- (iii) the median is smaller than the mean

- (c) Is the standard deviation of histogram (iii) around 5, 15, or 50?

around 15

- (d) True or false, and explain: the standard deviation for histogram (i) is a lot smaller than that for histogram (iii).

False. They are about the same.

2. Explain what the following Python code does. Iris is a DataFrame containing the iris dataset.

```
petal_measurements = iris[["petal.length", "petal.width"]]
```

This code selects the `petal.length` and `petal.width` columns from the `iris` DataFrame and stores them in a new DataFrame named `petal_measurements`. The double brackets indicate that a list of column names is being used, so the result contains both columns and remains a DataFrame.